Chae-Eun Yoon

PhD Student · Biomedical Engineering · UNIST

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Education

Ulsan National Institute of Science and Technology (UNIST) *Ulsan, Republic of Korea*
Integrated M.S.–Ph.D. in Biomedical Engineering *2023 – present*

- Brain–Computer Interface (BCI) Lab · Advisor: Sung-Phil Kim

Pusan National University *Busan, Republic of Korea*
B.S. in Chemical, Biological and Environmental Engineering *2019 – 2023*

Research Experience

Brain–Computer Interface Lab, UNIST *Ulsan, Republic of Korea*
Graduate Researcher *2023 – present*

- EEG decoding of musical pitch and melody imagery; closed-loop music-imagery BCIs.

Water Treatment and Reuse (WATER) Lab, Pusan National University *Busan, Republic of Korea*
Undergraduate Researcher *2021 – 2022*

Grants & Fellowships

Doctoral Student Research Grant, National Research Foundation of Korea (NRF) *2025 – 2027*
Principal Investigator

- Real-Time Playback of **Imagined Melodies** via a Closed-Loop Noninvasive EEG Brain–Computer Interface.

Core Research Program (Type B), National Research Foundation of Korea (NRF) *2025 – 2030*
Participating Researcher

- Music Braincoder: Development of a noninvasive brain–computer interface for reproducing **imagined music** and enhancing musical performance.

Publications

Kim, J., **Yoon, C.-E.**, Seo, S., & Kim, S.-P. (2026). Real-Time Single-Trial Decoding in a Calibration-Free P300-Based BCI Using a Compact Eight-Channel EEG Configuration. *2026 14th International Conference on Brain–Computer Interface (BCI)*, 1–6.

Conference Presentations

Yoon, C.-E., & Kim, S.-P. (2026). Decoding Melodic Contour from Human Electroencephalography during Music Listening, Imagery, and Production. *Society for Neuroscience (SfN)*. *(forthcoming)*

Yoon, C.-E., & Kim, S.-P. (2026). Decoding Melodic Contour from Human Electroencephalography Using Spectral Patterns. *Brain Engineering Society of Korea (BESK)*, Jeju, Republic of Korea. *(forthcoming)*

Yoon, C.-E., & Kim, S.-P. (2025). Effects of Visual Cues on Musical Melody Imagery for EEG-based Brain–Computer Interfaces. *Society for Neuroscience (SfN)*.

Yoon, C.-E., & Kim, S.-P. (2025). Neural Representation of Musical Pitch During Music Perception and Imagery. *International Society for Music Information Retrieval (ISMIR)*.

Yoon, C.-E., & Kim, S.-P. (2023). Design of Visual Guidance for Musical Pitch Imagery. *5th International Conference on Emotion and Sensibility (ICES)*, Jeju, Republic of Korea.

Teaching

Brain & Cognitive Engineering *Fall 2025*
Teaching Assistant — UNIST

AI-based Neural Data Science *Spring 2024*
Teaching Assistant — UNIST

Research Interests

- Music neuroscience; neural representation of pitch and melody
- EEG decoding & brain–computer interfaces (BCI)
- Music imagery and production
- Open, reproducible neuroscience

Last updated: June 2026